Evaluation Choices Determine Apparent Quantum Kernel Robustness under Distribution Shift

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Abstract

Claims of quantum-kernel robustness can depend on how candidates are tuned, selected, and compared. We isolate this dependence with a controlled kernel-swap benchmark under distribution shift, a ten-specification curve, and prospectively frozen corroboration on three external domain shifts. In eight fixed security scenario-groups, same-test target-label selection, fixed regularization, and weak classical references produce apparent gains up to $\mathbf{+0.037}$. Train-only regularization, target-label-free selection, and structurally matched candidate budgets instead give source-dataset-equal quantum-minus-classical effects of $\mathbf{-0.0056}$ for support-vector classification and $\mathbf{-0.0009}$ for Gaussian-process classification. Across the external tasks, the controlled support-vector effect is $\mathbf{-0.0119}$ (conditional 95% interval $[\mathbf{-0.0205, -0.0033}]$) and protocol contraction is $\mathbf{+0.0208}$. Effective rank and alignment associate with accuracy but occur in both families. Repeated finite-shot perturbations reveal measurement- and projection-dependent deviations without simulating hardware. The apparent robustness verdict is therefore evaluation-dependent in these low-qubit regimes.

Keywords: quantum machine learning; quantum kernels; distribution shift; out-of-distribution robustness; kernel methods; kernel-target alignment; Gaussian processes; malware detection; network intrusion detection

1 Introduction

Quantum kernel methods embed classical data into quantum states and use state overlaps as similarities for otherwise classical learners [1–3]. This separation makes them a clean setting for asking whether quantum-induced feature geometry provides useful inductive bias. It also makes comparison unusually sensitive to evaluation design: the downstream learner can be held fixed, but conclusions still depend on the encoding, bandwidth, regularization, classical reference set, and rule used to select a candidate [4–7].

Recent benchmarks have accordingly shifted the question from whether an isolated quantum model can perform well to whether a claimed gain survives strong and symmetric controls. Broad QML studies find competitive classical baselines once pipelines are aligned [6]; quantum-kernel benchmarks show substantial sensitivity to feature map and hyperparameter choice [7]; and a concurrent tabular study using nested cross-validation, spectral analysis, and hardware validation finds no significant quantum win across its tested pairs [8]. These studies evaluate predominantly identically distributed train–test settings. They do not address a further deployment question: what happens when the target distribution changes and its labels cannot be used to choose the deployed kernel?

Distribution shift makes model selection part of the estimand. Work on domain generalization and real-world shift has shown that robustness gains can disappear when search budgets, hyperparameters, and selection criteria are standardized [9–12]. A selector that inspects the shifted test labels estimates an oracle upper bound, not deployable performance; a family with more candidates has more opportunities to win that selection. A controlled quantum–classical evaluation under shift must therefore hold fixed not only the classifier and data split, but also regularization, candidate budget, and label access.

Security detection provides a useful stress test because shift is intrinsic to the application. Malware evolves over time, and experimental bias can overstate detector performance [13–15]; network traffic changes with infrastructure and attack campaigns [16, 17]. Quantum classifiers have already been evaluated on intrusion-detection benchmarks, sometimes with reported gains over small classical reference sets on i.i.d. splits [18–20]. We use these datasets not to propose a security system, but to expose kernel families to reproducible constructed shifts and to a held-out attack-campaign shift.

The closest robustness-specific concurrent work studies additive input corruption and proposes a spectral phase encoding [21]. Our question is complementary: rather than optimize a new encoding, we audit how evaluation choices change a family-level conclusion under constructed and held-out-campaign shifts. The distinguishing axes are no-OOD-label selection, equal candidate budgets, per-configuration train-only regularization, fixed-case inference, and a geometry analysis tied specifically to behaviour after the distribution moves.

We implement a controlled *kernel-swap* benchmark in which preprocessing, splits, classifier, and family-internal selection logic are fixed while only the kernel changes. Four security scenarios from EMBER, UNSW-NB15, and ToN-IoT yield eight scenario-groups across constructed and held-out-campaign shifts. Support-vector and Laplace Gaussian-process classifiers consume the same precomputed Gram matrices. The quantum family contains four fidelity feature-map families with angle-scale sweeps; the classical family extends linear and RBF references with polynomial, Laplacian, and Matérn kernels and symmetric bandwidth freedom. Each configuration's regularization is tuned from training data alone. The primary deployed candidate is selected on a disjoint ID-validation split, and the two extended families receive equal candidate budgets. Fifteen pipeline realizations per setting quantify conditional variability; the benchmarks are treated as fixed case studies rather than as random draws from an undefined population.

The comparison is coupled to a geometry audit. We measure effective rank, centered kernel-target alignment, concentration, and geometric difference [4, 22–24]; use partial association, leave-one-source-dataset-out prediction, and effective-rank matching to restrict interpretation; and repeat finite-shot perturbations to quantify measurement-sampling variability before and after positive-semidefinite projection.

The evaluation choices change the conclusion. With same-test oracle selection, fixed regularization, and customary linear+RBF baselines, fidelity kernels appear to improve OOD balanced accuracy by as much as +0.037. With train-only regularization, no-OOD-label selection, and structurally matched candidate budgets, the source-dataset-equal quantum-minus-classical delta is −0.0056 for SVC and −0.0009 for the GP classifier. The primary SVC comparison is classical-favoured or conditionally tied in seven of eight scenario-groups. Effective rank and OOD alignment remain associated with accuracy, but the association is regime-dependent and shared by both families. Nearest-rank pairing within the predefined overlap yields negative median differences in all eight groups, including under one-to-one and caliper sensitivities. Repeated finite-shot estimates inflate exact-statevector effective rank and produce condition-dependent OOD deviations.

The contribution is therefore the combined evaluation design: distribution shift, target-label-free selection, train-only regularization, structurally matched search budgets, a prespecified specification curve, and prospectively frozen external corroboration. Together these controls identify which choices create the optimistic quantum-kernel robustness result and show that the observed geometry does not uniquely identify quantum provenance. The scope is deliberately limited to the low-qubit fidelity kernels, datasets, and shift constructions studied here.

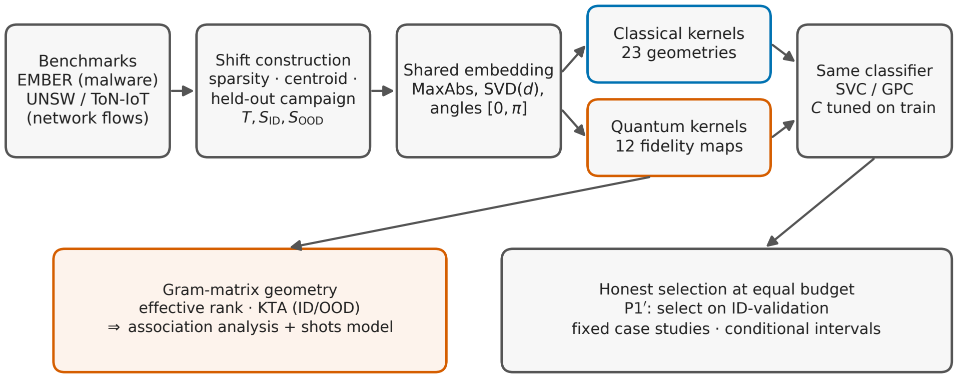

Fig. 1 The controlled kernel-swap protocol. Benchmark pools feed reproducible shift constructions; all kernels operate on a shared train-only embedding, receive per-configuration regularization tuned on the training split, and are consumed by the same precomputed-kernel classifiers. Family-level conclusions are drawn by target-label-free, equal-budget selection on an in-distribution validation split (P1′). Each scenario-group interval is a 95% conditional cluster-t interval over five q-split seed clusters ($n = 5$), conditional on the fixed benchmark pools and design. Gram matrices feed the associational geometry audit and repeated finite-shot sensitivity.

2 Results

2.1 Overview

We report results under the controlled kernel-swap protocol described in Section 4. All comparisons share the same classifier, preprocessing pipeline, and ID/OOD split definitions within a scenario-group; the only varying component is the kernel. Each setting contains fifteen pipeline realizations arranged in five q-split seed clusters, not fifteen independent observations. The argument proceeds in three movements. First, under test-peeking oracle selection and fixed regularization, the fidelity kernels show an apparent OOD advantage over linear and RBF baselines (Section 2.2). Second, we tune each configuration on training data alone, select on an in-distribution validation split disjoint from both test splits, and match both the count and block structure of the extended-family search budgets. Under this controlled comparison no robust family-level advantage remains; the primary SVC comparison is classical-favoured or tied in seven of eight scenario-groups (Section 2.4). Third, a geometry analysis shows that effective rank and OOD alignment are associated with—but do not causally determine—performance, that the association is regime-dependent, and that it favours neither family (Section 2.6); repeated finite-shot perturbations then quantify how sampling and positive-semidefinite projection alter exact-statevector endpoints (Section 2.7). Throughout, the eight scenario-groups are fixed case studies; conditional intervals describe pipeline variation given the benchmark pools and design, not uncertainty over a population of datasets.

2.2 The apparent advantage under oracle selection

We begin with the comparison that dominates the literature: fidelity-based quantum kernels against linear and RBF baselines, with each family's representative chosen by Best-by-OOD selection—the configuration with the highest balanced accuracy on the OOD test split itself—and the classical kernels left at the default regularization $C = 1$. Under these conventions, on EMBER, the quantum family shows an OOD advantage of +0.037 under SVC and +0.028 under the GP classifier against linear+RBF, shrinking to +0.004/+0.002 once the classical family is enlarged with heavier-tailed kernels. Supplementary Table 5 gives the complete oracle results.

Two features favour this apparent quantum result. Best-by-OOD is an *oracle*: it uses the labels it then reports. Fixing $C = 1$ also leaves classical candidates at a potentially unsuitable regularization point. These values are therefore descriptive upper bounds, not deployable effects.

2.3 A specification curve locates the reversal

To separate the contribution of each evaluation choice from the choice of a preferred endpoint, we enumerated all ten logically valid, already-computed specifications fixed before this extension (Figure 2). The specifications vary four axes in a prespecified order: same-test OOD-oracle versus ID-based selection, fixed versus train-cross-validated SVC regularization, customary linear/RBF versus extended classical references, and native versus equal candidate budgets. Every specification retains all eight security scenario-groups.

The family-level conclusion spans both sides of zero. For SVC, the source-dataset-equal quantum-minus-classical mean ranges from $+0.047$ under legacy ID selection with fixed $C = 1$ and the customary reference (S2) to -0.006 under ID-validation selection, train-only regularization, the extended reference, and equal budgets (S10). For the GP classifier, for which the regularization axis is not applicable, the corresponding range is $+0.042$ to -0.003. Expanding the classical reference contracts the positive effect before target-label-free selection or budget matching is imposed; train-only SVC regularization and ID-validation selection then move the controlled estimates to zero or slightly negative. The path is not monotone in every intermediate specification, which is precisely why reporting only an oracle and a final controlled endpoint would be insufficient. The curve shows that the apparent advantage is conditional on a bundle of evaluation decisions rather than a stable property of quantum provenance.

2.4 The controlled comparison removes the apparent advantage

A deployable comparison must tune each configuration's regularization without looking at the test data, select the deployed configuration without OOD labels, and compare families at the same candidate budget. We implement all three (Section 4). For every kernel candidate, the SVC regularization C is chosen by five-fold cross-validation on the training Gram matrix alone. The deployed configuration is then the one with the best balanced accuracy on an in-distribution *validation* split, held out from the ID test split (protocol P1′); no OOD label is ever consulted in selection. The primary comparison uses the same candidate budget for both extended families (60 per family in full-coverage groups and 20 in reduced-coverage groups), with

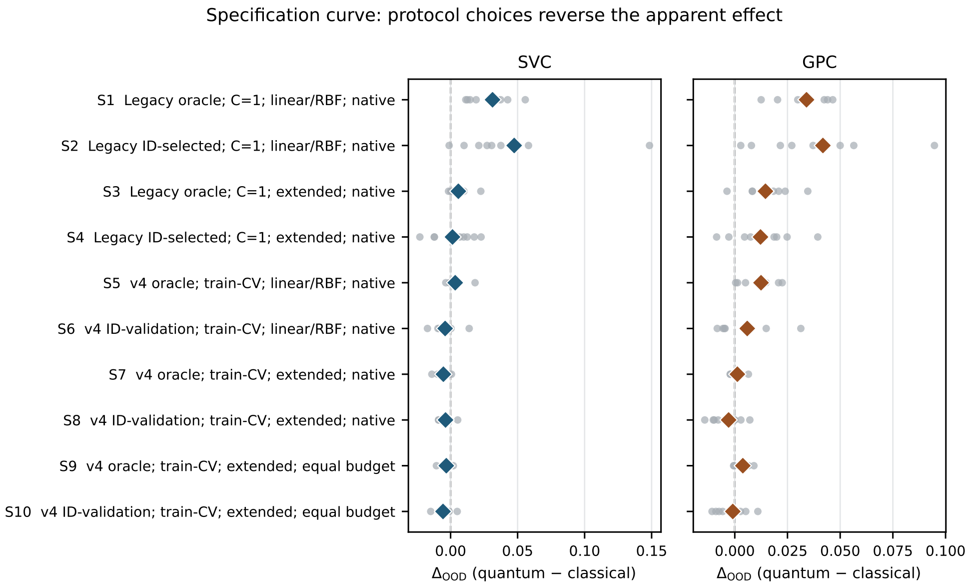

Fig. 2 A frozen specification curve shows that evaluation choices reverse the apparent family effect. Grey points are the eight fixed security scenario-groups; diamonds average groups within EMBER, UNSW-NB15, and ToN-IoT before giving the three source datasets equal weight. Specifications are shown in their prespecified S1–S10 order, not sorted by effect. They vary target-label access, SVC regularization, classical reference strength, and candidate budget. Positive values favour the fidelity-kernel family. GPC has no C-regularization axis. These are descriptive fixed-case summaries; no specification-level hypothesis test or population interval is reported.

the full classical pool retained as a sensitivity. We treat the eight scenario-groups as fixed case studies and report, per group, the quantum-minus-classical OOD delta with a conditional interval over the five split realizations, and no population p-value (Section 4).

Table 1 and Figure 3 give the result: no robust advantage remains against either reference. Against the equal-budget extended classical family, the P1′ delta is classical-favoured or within the conditional uncertainty of zero in seven of the eight scenario-groups under the primary SVC comparison. Averaging groups within each source dataset and then giving EMBER, UNSW-NB15, and ToN-IoT equal weight gives -0.00565 (leave-one-source-dataset-out range $[-0.00702, -0.00450]$); the GP-classifier mean is -0.00094 ($[-0.00190, +0.00094]$). Against the customary

Table 1 Target-label-free selection results for eight fixed scenario-groups. Values are quantum-minus-classical OOD balanced-accuracy differences under P1′ (select on ID-validation, tune per-configuration C on train, evaluate on OOD test). The extended-family comparison is the frozen equal-budget analysis: 60 candidates per family in full-coverage groups and 20 in reduced-coverage groups. Each scenario interval is a 95% conditional cluster-t interval over five q-split seed clusters ($n = 5$), conditional on the benchmark pools and design; the fifteen pipeline realizations per setting are nested within these clusters. Source-dataset-equal rows average groups within three source datasets and report leave-one-source-dataset-out ranges, not confidence intervals. The customary linear+RBF comparison is contextual rather than equal-budget.

Scenario	Clf.	vs lin./RBF	vs ext. (equal B)	conditional interval
EMBER $m1$	SVC	−0.0095	−0.0058	[−0.0340, +0.0223]
	GPC	−0.0050	−0.0009	[−0.0101, +0.0083]
EMBER $m2$	SVC	+0.0139	−0.0064	[−0.0137, +0.0009]
	GPC	+0.0149	+0.0027	[−0.0027, +0.0081]
UNSW-DoS (campaign)	SVC	−0.0043	−0.0033	[−0.0139, +0.0073]
	GPC	+0.0041	−0.0057	[−0.0120, +0.0005]
UNSW-DoS (constructed)	SVC	−0.0085	−0.0044	[−0.0119, +0.0031]
	GPC	−0.0056	−0.0108	[−0.0177, −0.0039]
UNSW-Recon (campaign)	SVC	+0.0004	+0.0050	[−0.0058, +0.0157]
	GPC	+0.0074	+0.0052	[−0.0019, +0.0123]
UNSW-Recon (constructed)	SVC	−0.0076	−0.0089	[−0.0146, −0.0031]
	GPC	−0.0083	−0.0075	[−0.0208, +0.0059]
ToN-IoT (campaign)	SVC	−0.0172	−0.0148	[−0.0268, −0.0029]
	GPC	−0.0046	−0.0090	[−0.0190, +0.0010]
ToN-IoT (constructed)	SVC	−0.0016	−0.0011	[−0.0049, +0.0028]
	GPC	+0.0313	+0.0109	[+0.0017, +0.0202]
Source-dataset-equal mean, SVC		−0.0041	−0.0056	[−0.0070, −0.0045]
Source-dataset-equal mean, GPC		+0.0059	−0.0009	[−0.0019, +0.0009]

linear+RBF baseline—the comparison that produced the +0.037 oracle result of Section 2.2—the target-label-free delta is −0.00407 under SVC and +0.00591 under the GP classifier, with heterogeneous signs across scenario-groups. The single regime that retains a small quantum lean under both equal-budget comparisons is UNSW-NB15 reconnaissance under held-out attack-campaign shift (+0.005 for both classifiers); both conditional intervals include zero. The full-pool extended-family sensitivity is similar in direction and scale (source-dataset-equal means −0.00378 for SVC and −0.00293 for GPC), so the conclusion does not depend on giving the classical family its larger pool.

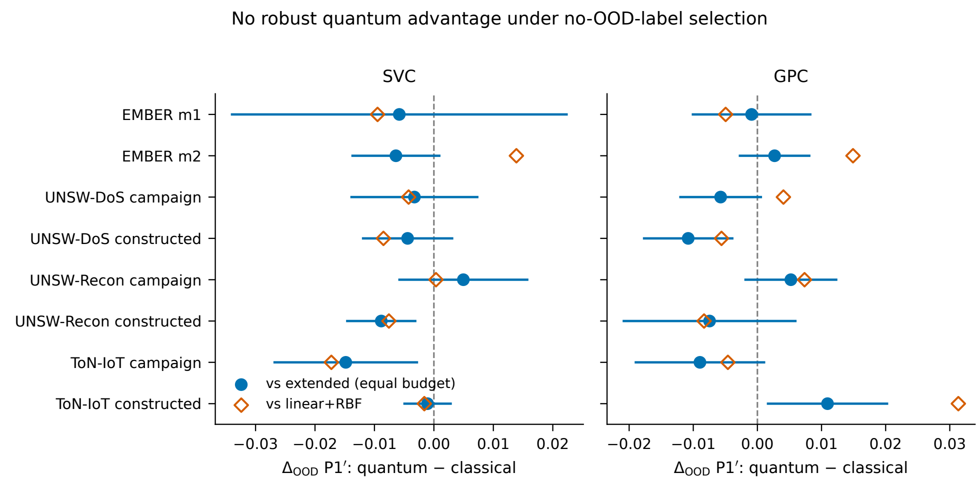

Fig. 3 No robust quantum advantage under target-label-free selection. Per-scenario-group P1′ OOD differences (quantum minus classical) are shown for SVC (left) and the GP classifier (right). Filled circles compare against the equal-budget extended classical family; their error bars are 95% conditional cluster-t intervals over five q-split seed clusters ($n = 5$), conditional on the fixed benchmark pools and design. Open diamonds compare against the customary linear+RBF reference. All eight fixed scenario-groups are shown; no population interval is implied.

The candidate-budget result is insensitive to the sampling scheme.

The primary `kernel_blocked` scheme samples 12 whole classical kernel-shape blocks crossed with all five dimensions, matching both the quantum count and its feature-map-by-dimension structure. Uniform and kernel-stratified samples match the count but not this block structure. The source-dataset-equal resampling estimates remain close under all three schemes: −0.00576, −0.00555, and −0.00548 for SVC, and −0.00185, −0.00232, and −0.00253 for GPC, respectively. The largest scheme range in any group–classifier cell is 0.00353. Supplementary Table 2 reports all 48 cells. Candidate-count equality is therefore distinct from search-space isomorphism, but this distinction does not drive the conclusion here.

Rank matching does not reveal a residual quantum advantage.

We pair each quantum configuration with the classical configuration of nearest training effective rank within the same run and dimension. This observational matching reuses

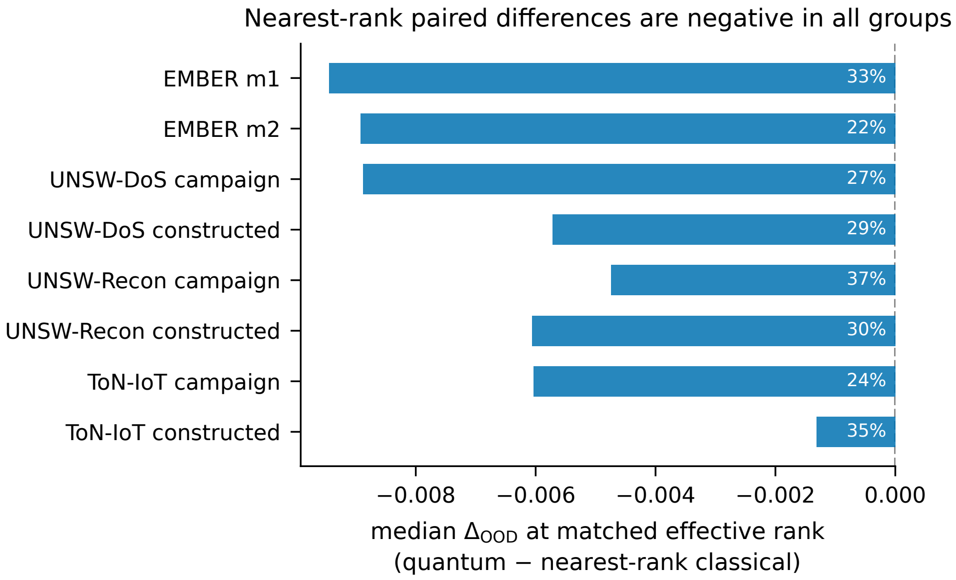

Fig. 4 Nearest-rank pairing yields negative median quantum-minus-classical differences in all eight fixed scenario-groups. Each quantum configuration is paired, with replacement, to the nearest-effective-rank classical configuration within run and dimension, subject to rank ratio ≤ 1.25. The filter retains 48,708/64,800 pairs (75.2%; 69.3–83.3% by group). Bars show per-group median OOD balanced-accuracy differences; annotations give the fraction of retained pairs in which the quantum member is more accurate. This is observational matching; classical candidates may be reused and no causal family effect is implied.

classical candidates and applies the prespecified caliper $\max(r_Q, r_C)/\min(r_Q, r_C) \leq 1.25$. It retains 48,708 of 64,800 pairs (75.2%; 69.3–83.3% by group). Figure 4 reports negative median quantum-minus-classical differences in all eight groups (-0.0013 to -0.0094); the quantum member is more accurate in 22–37% of retained pairs. The pooled median is -0.0056. Minimum-cost one-to-one matching retains 24,675 pairs at the same caliper and yields a pooled median of -0.0053, again negative in every group. Results remain negative over calipers 1.10, 1.50, 2.00, and without a caliper (Supplementary Table 3). These pairings describe the predefined region of rank overlap; they do not identify a causal effect of kernel family.

2.5 Protocol sensitivity is prospectively corroborated on external domain shifts

We next applied the prospectively frozen protocol to College Scorecard, diabetes readmission, and ACS income, preserving the published TableShift domain splits and withholding all OOD labels from the controlled selector. The complete external grid contains 30 task–size–seed units and 37,800 split-level results; every prespecified unit passed the independent audit. Figure 5 shows all task–size effects and conditional seed-cluster intervals.

The primary SVC result satisfies the machine-readable outcome category frozen as `replicated_protocol_sensitivity`; narratively, protocol sensitivity is prospectively corroborated across three fixed external tasks. The task-equal controlled effect is -0.0119 (95% conditional seed-cluster interval $[-0.0205, -0.0033]$, $n = 5$ deterministic subsampling seeds per task), whereas the weak-reference, fixed-C, same-test-oracle effect is $+0.0089$ ($[+0.0047, +0.0137]$). Their contraction is $+0.0208$ ($[+0.0113, +0.0307]$), with a leave-one-task-out range of $[+0.0116, +0.0260]$. No task has a positive controlled SVC effect whose interval excludes zero in both sizes. Size-equal controlled effects are -0.0160 for College Scorecard, -0.0066 for diabetes readmission, and -0.0131 for ACS income.

The secondary GP classifier shows a larger contraction of $+0.0324$ ($[+0.0223, +0.0427]$), but its task-equal controlled effect is $+0.0077$ with an interval spanning zero ($[-0.0015, +0.0183]$). College Scorecard leans quantum under this classifier, diabetes leans classical, and ACS is slightly positive. This classifier dependence limits the conclusion appropriately: the external study corroborates the sensitivity of the family verdict to evaluation design, not universal classical dominance.

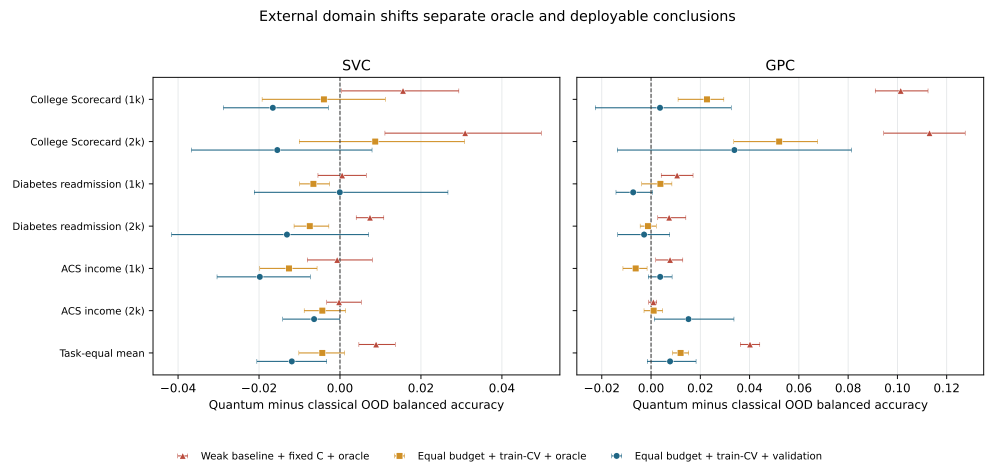

Fig. 5 External domain shifts separate optimistic and deployable conclusions. Effects are quantum minus classical OOD balanced accuracy for the weak-reference fixed-regularization same-test oracle, the equal-budget train-tuned same-test oracle, and the primary equal-budget train-tuned validation selector. Rows show both nested sizes for all three fixed tasks plus the task-equal mean. Error bars are 95% conditional seed-cluster intervals over five deterministic subsampling seeds per task ($n = 5$), preserving the nested-size relation. They quantify pipeline variation conditional on the fixed tasks and design, not uncertainty over a population of tasks.

2.6 Geometry is associated with, but does not determine, robustness

The results are consistent with both families reaching similar informative geometries. Geometry carries partial information about behaviour under shift, but neither the analysis nor the matching identifies it as a causal mediator: the association is regime-dependent, not the single governing law that optimistic accounts (our own earlier report included) proposed.

The geometry–OOD link is a regime-dependent association, not a law.

Two descriptors of the training and OOD Gram matrices carry information about which kernels generalize: the effective rank of the training kernel and the centered kernel-target alignment on the OOD split. But their association with OOD accuracy is regime-dependent, not a single mechanism. Within scenario-groups (Figure 6 and Supplementary Table 4), the median Spearman correlation between effective rank

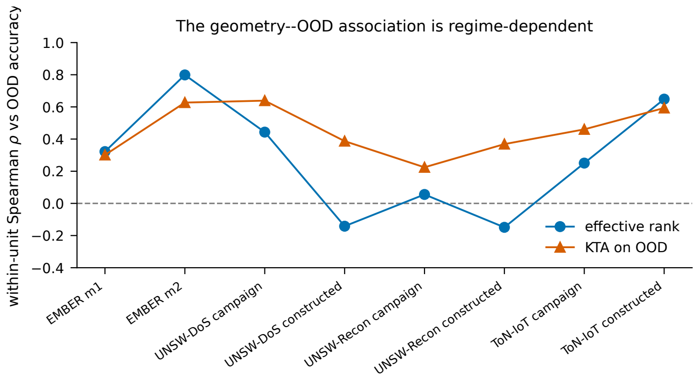

Fig. 6 The geometry–OOD association is regime-dependent. Points are median within-unit Spearman correlations between each geometry descriptor and OOD balanced accuracy, per fixed scenario-group for SVC; each unit is a run–dimension cell. Effective rank orders accuracy strongly in some regimes (EMBER $m2$, ToN-IoT $m2$) and weakly or negatively in others (the UNSW constructed-shift scenarios); OOD alignment is more consistently positive but never uniform. These descriptive associations have no population interval and do not identify a causal mechanism.

and OOD accuracy ranges from +0.80 (EMBER $m2$) down to −0.15 (UNSW-NB15 reconnaissance under the constructed shift); partial correlations that control for family label and length scale track the raw values closely. OOD alignment is more consistently positive (+0.22 to +0.64) but likewise varies by regime. A geometry model trained on two source datasets and used to rank kernels in the third achieves only Spearman 0.22–0.40 (EMBER 0.28, UNSW-NB15 0.22, ToN-IoT 0.40). We therefore describe a genuine but partial association, not a causal mechanism or deployable predictor.

2.7 Finite-shot estimation: the geometry is a statevector idealization

All quantum kernels above are computed by exact statevector simulation, which isolates kernel geometry but is not what a device would return. We therefore froze a post-confirmatory Monte Carlo sensitivity before computing its outcomes. The subset contains one $q1000$ run per scenario-group, all at master, q-split, and model seed 42; within each run, the exact P1′-selected quantum SVC configuration is fixed. For each of four shot counts, 30 independent measurement replicates replace every fidelity entry by a binomial estimate, reselect C from training data, and evaluate both the sampled matrix and its positive-semidefinite (PSD) projection. Stable SHA-256 seeds are distinct by run, configuration, shot count, replicate, and matrix block (Methods).

Table 2 summarizes the eight conditional Monte Carlo distributions. Sampling does not induce a uniform accuracy direction. After PSD projection, the median across the eight fixed scenario-group medians is -0.0005 at 128 shots (group range $[-0.0252, +0.0131]$), -0.0051 at 512, -0.0052 at 2,048, and 0.0000 at 8,192 shots. Before projection the corresponding values are -0.0044, -0.0009, -0.0005, and 0.0000. Thus PSD projection is part of the estimand rather than an innocuous cleanup step: it can change the sign and magnitude of an individual fixed-run deviation. Central 95% Monte Carlo intervals for all runs are shown in Supplementary Figure 2.

Effective rank remains more sensitive than OOD alignment. The median perturbed-to-exact effective-rank ratio falls from 1.93 at 128 shots to 1.13 at 8,192, while the across-group median OOD-alignment change is within 0.0007 of zero at every shot count. Median absolute OOD-accuracy deviations remain 0.0070–0.0084 after projection and are not monotone because each perturbed training matrix reselects a discrete C. These distributions quantify measurement-sampling variability conditional on eight fixed exact Gram matrices. They do not model a device, circuit compilation, gate or readout noise, mitigation, or hardware feasibility.

2.8 Predictive uncertainty under shift

The Laplace GP probabilities provide a secondary uncertainty check (Supplementary Figure 1). For the target-label-free selected candidate, predictive entropy rises from ID to OOD for both families in every scenario-group, while expected calibration error changes moderately and comparably. Because neither family has an accuracy advantage, the fidelity kernels show no calibration benefit unavailable to classical kernels. We do not interpret rising entropy alone as evidence of calibration.

Table 2 Repeated finite-shot fidelity-estimation sensitivity for eight fixed exact Gram matrices (SVC). Each row summarizes, across scenario-groups, the within-run median over 30 measurement replicates. Δ_{OOD} is perturbed minus exact OOD balanced accuracy; brackets give the range of the eight medians, not a confidence interval. $|\Delta_{\mathrm{OOD}}|$ is the across-group median absolute difference. Rank ratio is perturbed/exact effective rank after clipping negative spectral mass; pre- and post-PSD values coincide for this descriptor. Per-run central 95% Monte Carlo intervals are in Supplementary Figure 2.

Shots	Matrix	median Δ_{OOD} [group range]	median $\lvert\Delta_{\mathrm{OOD}}\rvert$	rank ratio	median Δ_{KTA}
128	pre-PSD	-0.0044 $[-0.0563, +0.0137]$	0.0118	1.93	-0.00071
128	post-PSD	-0.0005 $[-0.0252, +0.0131]$	0.0070	1.93	-0.00037
512	pre-PSD	-0.0009 $[-0.0226, +0.0192]$	0.0068	1.48	-0.00017
512	post-PSD	-0.0051 $[-0.0281, +0.0041]$	0.0084	1.48	-0.00006
2,048	pre-PSD	-0.0005 $[-0.0222, +0.0055]$	0.0080	1.24	-0.00005
2,048	post-PSD	-0.0052 $[-0.0238, +0.0041]$	0.0072	1.24	$+0.00001$
8,192	pre-PSD	0.0000 $[-0.0158, +0.0114]$	0.0101	1.13	-0.00001
8,192	post-PSD	0.0000 $[-0.0149, +0.0143]$	0.0081	1.13	$+0.00001$

3 Discussion

The study's central interpretive move is to replace the question "do quantum kernels beat classical kernels under shift?" with "which properties of a kernel are associated with its behaviour under shift, and does either family have exclusive access to them?". Read through that lens, the results support a narrower conclusion: the apparent advantage is evaluation-sensitive. It does not survive the combined control of target-label access, train-only regularization, and candidate search, does not reappear against the equal-budget extended family, and nearest-rank pairing yields negative median differences throughout the predefined overlap. The geometry descriptors carry information as regime-dependent associations rather than a governing law, and they are shared: well-tuned heavy-tailed classical kernels occupy observed geometric regimes also reached by fidelity maps. On present evidence there is no robust out-of-distribution advantage for fidelity kernels in the low-qubit settings studied. A modest in-distribution separability edge under the probabilistic classifier remains, but is small, classifier-dependent, and shrinks under target-label-free selection.

From the perspective of the distribution-shift literature, this reinforces a general lesson: conclusions about robustness are conditional on the protocol used to obtain them [9–11, 25]. Our results add a QML-specific corollary. The strength of the classical baseline, whether regularization is tuned, whether model selection may consult target labels, and the structure of each family's candidate search can each move the quantum-versus-classical estimate. Reporting these choices is therefore necessary for interpreting an apparent +0.037 advantage or comparing it with a controlled endpoint near zero.

The external analysis provides prospective corroboration beyond the constructed security shifts. Under a protocol frozen before result inspection, three education, health, and socioeconomic tasks corroborate a positive protocol contraction and a negative controlled SVC effect. The result is stable to omitting any one task, yet heterogeneous by task and classifier: the GP endpoint retains a small positive task-equal lean whose interval crosses zero. The strongest defensible conclusion is therefore about evaluation sensitivity, not the universal ordering of kernel families.

In-distribution separability is the one axis on which the fidelity kernels retain an edge, and even it must be stated carefully. Under the oracle that selects and evaluates on the same in-distribution split, the quantum family separates the classes slightly better than linear+RBF in every EMBER setting; but under target-label-free selection—choosing on a validation split and reporting on a disjoint test split—the edge falls to +0.003 under SVC (mixed in sign across groups) and +0.012 under the probabilistic classifier, and it does not translate into an observed out-of-distribution benefit. This modest ID–OOD dissociation does not by itself establish operational value for shift-sensitive deployment. The pattern is consistent with the "accuracy on the line" literature [26], in which ID and OOD accuracy usually move together; here they decouple weakly, while descriptors shared by both families mark part of the axis along which they do.

The geometry association connects several strands of literature. That length scale is decisive for kernel behaviour is known for both families [27, 28]; our comparative result is that, once length scale is tuned symmetrically and regularization is tuned, the two families reach overlapping geometric regimes, so asymmetric tuning can change the family comparison. The geometric difference of Huang et al. [4] does not predict which family wins a scenario, consistent with its definition as a bound on the *potential* for a separation rather than its sign; a large geometric difference is compatible with equal or worse OOD accuracy, as we observe. Effective rank and OOD alignment are therefore descriptors of kernels in general, not unique signatures of quantum provenance.

The high-rank finding must also be reconciled with two spectral results that superficially point the other way. Kübler et al. argue that quantum kernels generalize in-distribution only when the relevant spectral mass is concentrated in few components [29], and Thanasilp et al. show that fidelity kernels concentrate exponentially—toward effectively rank-one, uninformative Gram matrices—as qubit counts grow [30]. There is no contradiction, but there is a genuine tension that our results locate precisely. Both results concern the asymptotic or i.i.d. regime; our effective ranks (roughly 10–150 on training sets of 10^3–4×10^3 samples at 4–12 qubits) sit far below the concentration regime. The repeated finite-shot sensitivity adds an empirical caveat these theoretical results anticipate: effective rank from sampled fidelities is inflated by a median factor of 1.93 at 128 shots and 1.13 at 8,192 shots (Section 2.7). Exact effective rank is therefore a statevector idealization even at these modest qubit counts.

We emphasize what the geometry analysis does not deliver. It is a post-hoc association, not a deployable selection rule: choosing a configuration by training-set geometry alone does not reliably identify the better kernel, and OOD alignment requires OOD labels. Because the controlled comparisons do not yield a consistent quantum advantage under target-label-free selection, a more useful open problem is whether any label-free, train-only signal can select a robust kernel at all—a question that concerns classical and quantum kernels equally.

From a practical perspective, the paper suggests three takeaways for security ML. The first is substantive: on the evidence here, fidelity-based quantum kernels do not offer a robustness advantage under distribution shift over well-tuned classical kernels, while their pairwise overlap estimation introduces circuit and shot costs absent from direct evaluation of the classical baselines. A median-heuristic Laplacian kernel, especially one with a shortened length scale and its regularization tuned, is a strong and inexpensive robustness baseline, and it matches the fidelity maps under no-OOD-label selection on both modalities. Practitioners weighing a quantum-kernel pipeline for OOD-sensitive security detection should, on this evidence, prefer a tuned classical kernel unless a specific regime demonstrably favours the quantum construction after the same controls.

The second is methodological. Independently of the preferred kernel family, the protocol used here provides a reusable blueprint for robust evaluation in security settings:

1. keep the classifier and preprocessing fixed, and swap only the kernel;
2. define explicit OOD splits tied to interpretable shift mechanisms, including at least one non-constructed shift mechanism where the data permit;
3. give every family the same freedom—the same candidate budget, symmetric length-scale tuning, and per-configuration regularization tuned on training data alone;
4. select the deployed configuration without out-of-distribution labels, and keep an in-distribution test split disjoint from the validation split used for selection;
5. repeat the pipeline, report variability as conditional intervals, and treat correlated benchmarks as fixed case studies rather than pooling them into a single population p-value.

Security ML frequently suffers from comparisons in which model class, tuning effort, and evaluation conditions change simultaneously; the way the estimate of Sections 2.2–2.4 moves from a large apparent advantage to a near-zero or negative endpoint as controls are added demonstrates how strongly the conclusion depends on the evaluation design.

The third concerns diagnostics: the geometry descriptors are cheap to compute from Gram matrices that the pipeline already builds, and the alignment-survival analysis can be run post hoc on any deployed kernel system with a labeled drift sample. Practitioners should nonetheless resist reading these results as a blanket recommendation to replace classical kernels; the responsible recommendation is to test the candidate families under the expected shift regime, using exactly the kind of protocol described here.

Several limitations define the scope of our conclusions.

Our shift operationalization covers three reproducible mechanisms—within-class sparsity-tail extremes, train-geometry-tail extremes, and a held-out attack-campaign shift on network traffic (an unseen attack campaign plus a capture-partition change, not a timestamped temporal split)—but not the full space of real-world drift processes. Explicit temporal splits on timestamped malware, family-based drift, and adversarially induced shift remain outside the study's scope, and our conclusions should be re-tested under them rather than assumed.

Kernel and encoding coverage also remains finite. The classical family spans seven kernels plus a bandwidth sweep, and both families receive symmetric tuning freedom, but neither candidate set is exhaustive: the quantum side is restricted to four commonly used fidelity feature maps at low qubit counts, and trained or task-adapted encodings, projected kernels, and deeper maps are not evaluated. The observed association suggests that future extensions should report their Gram-matrix geometry as well as provenance, but it does not predict their outcome. At substantially larger qubit counts, exponential kernel concentration [30] creates an additional regime not represented here, so our conclusions should not be extrapolated beyond the low-qubit study.

Selection is necessarily over finite candidate sets. P1′ (ID-validation selection), P2 (cross-realization OOD-supervised selection), and P3 (same-test OOD oracle) are selection procedures over finite candidate sets. The primary comparison matches candidate budgets, and the full-pool and budget curves provide sensitivities, but residual dependence on candidate definitions and search granularity remains. Fully nested selection over additional independent benchmark pools would strengthen the analysis further.

The quantum kernels are simulated rather than executed on hardware. The study therefore characterizes fidelity-kernel geometry, not noisy-device behaviour. Repeated binomial sampling quantifies one component of measurement uncertainty, but hardware noise, transpilation, readout, mitigation, and circuit-dependent correlations remain unmodelled [31]; exponential concentration may also make informative spectral structure expensive to resolve at larger scales [30]. We make no claim about hardware feasibility, resource advantage, or quantum speedup—the only rigorously established quantum-classical learning separation concerns a specifically constructed problem, not natural data [32]. At the dimensions studied, the fidelity kernels are also classically simulable [33, 34], which is precisely what makes them available as practical kernel constructions today.

Dataset scope and feature caveats provide a final boundary. Four benchmark scenarios across two modalities is a substantial widening relative to single-dataset studies, but security is broader still: dynamic malware analysis and encrypted traffic remain untested. Within the network scenarios, the feature sets inherited from the public benchmarks retain port numbers and protocol indicators, which are legitimate but known shortcut-prone features in NIDS benchmarks [35]; because every kernel receives identical features, this affects absolute accuracies, not the family comparisons. Finally, the mechanism's effective-rank component is itself regime-dependent (strong on EMBER and ToN-IoT, weak on UNSW-Reconnaissance), a gradation we report rather than smooth over.

In light of these limitations, the appropriate interpretation of the paper is deliberately conservative. We do not claim universal superiority, complexity-theoretic quantum advantage, or hardware relevance—nor do we claim to have refuted quantum kernels in general. We claim three narrower things, supported by conditional effect estimates rather than a population p-value: that the apparent out-of-distribution advantage of fidelity kernels over customary baselines disappears when same-test oracle selection and untuned regularization are removed; that under equal candidate budgets and no-OOD-label selection no robust family-level advantage remains against either the customary or the extended classical reference, across eight fixed scenario-groups and two classifiers; and that the geometric descriptors associated with out-of-distribution behaviour occur in both families and do not identify kernel provenance as the source of robustness. This view is consistent with recent calls for careful benchmarking in QML [6–8] and is more informative than either an unqualified advantage claim or an unqualified dismissal: it identifies the evaluation choices that generate the optimistic result and shows how the conclusion changes when they are controlled.

This paper asked why kernel-family verdicts change under distribution shift, using the quantum-versus-classical comparison as both motivation and stress test. A controlled kernel-swap protocol—classifier, preprocessing, splits, selection logic, and tuning freedom held fixed while only the kernel varies—was instantiated on four security scenarios spanning two modalities and three shift mechanisms, two classifier families including a probabilistic one, and 72 principal settings. Each setting contains fifteen pipeline realizations nested in five effective q-split clusters. Effects are fixed-case summaries with conditional cluster intervals, not population significance tests.

The empirical arc has three movements. Under same-test oracle selection and fixed regularization, fidelity-based quantum kernels appear to beat the customary linear and RBF baselines under shift. That result does not survive the controlled comparison: once each configuration is tuned on training data, selection uses no OOD labels, and both extended families search a structurally matched candidate budget, the source-dataset-equal OOD difference is -0.0056 under the primary SVC analysis and -0.0009 under the GP classifier. The primary comparison is classical-favoured or tied in seven of eight scenario-groups. Nearest-rank pairing yields negative median differences in every group under with-replacement, one-to-one, and caliper sensitivities. Effective rank and OOD alignment associate with performance in a regime-dependent way, occur in both families, and change under repeated finite-shot estimation.

The central contribution is the combined protocol: distribution shift, target-label-free selection, train-only regularization, structurally matched search budgets, a prespecified specification curve, and prospectively frozen external corroboration. Empirically, no robust family-level advantage survives these controls in the studied regimes. Methodologically, the study isolates the evaluation choices associated with the optimistic endpoint and provides a reproducible protocol for controlling each one. Analytically, it shows that geometric quantities invoked to explain such advantages describe overlapping regions of both families and change under finite-shot estimation.

The most useful open problem this leaves is not "which family wins" but whether any label-free, train-only signal can select a robust kernel under shift at all; the security community's label-free drift detectors [36] and the theory of task–model alignment [37–39] are natural starting points. Answering it would benefit classical and quantum methods alike: the operative question is which similarity geometry aligns with the shift regime one actually faces, and how to identify it without target-label access.

4 Methods

4.1 Problem setting and estimands

We study binary security classification—malware versus benign software and attack versus benign traffic—under distribution shift. Let D_{ID} denote the in-distribution process and D_{OOD} a constructed stress-test or held-out-campaign process. Each run contains a training set T, an ID holdout that is divided into validation V_{ID} and test S_{ID}, and an OOD test set S_{OOD}. Preprocessing, samples, classifier implementation, tuning rule, and family-selection rule are shared; only the kernel changes. Comparisons are always within a classifier.

The primary endpoint is OOD balanced accuracy after P1′ deployment selection. A family effect is

$$\Delta_{\mathrm{OOD}} = \mathrm{BAcc}_{\mathrm{OOD}}^{(Q)} - \mathrm{BAcc}_{\mathrm{OOD}}^{(C)},$$

so positive values favour the quantum family. ID balanced accuracy and ID-to-OOD degradation are secondary. P2 selects from OOD information in other realizations, and P3 selects and evaluates on the same OOD test labels; both are explicitly non-deployable descriptive oracles.

4.2 Security datasets and shared split sizes

The source datasets are EMBER [13], UNSW-NB15 [16], and ToN-IoT [17]. Four binary scenarios—EMBER malware, UNSW-NB15 DoS, UNSW-NB15 Reconnaissance, and ToN-IoT Scanning—produce two shift mechanisms each and hence eight scenario-groups. Classical and quantum kernels always receive identical row indices and class balance.

The 72 principal settings cross eight groups with master seeds $\{42, 123, 999\}$ and $(n_{\mathrm{train}}, n_{\mathrm{ID}}, n_{\mathrm{OOD}})$ equal to $(1000, 500, 500)$, $(2000, 1000, 1000)$, or $(4000, 1800, 1800)$. Each setting is instantiated with q-split seeds $\{42, 123, 999, 7, 2024\}$ and model seeds $\{42, 123, 999\}$, yielding fifteen pipeline realizations arranged in five q-split clusters. Indices are stored explicitly; automated gates verify split-role disjointness, required cardinalities, both classes, identical family inputs, and absence of label-derived features.

4.3 EMBER shift construction

EMBER uses the histogram and byte-entropy feature blocks and two label-conditional tail stress tests. They are not described as certified covariate shifts because conditioning the split on class does not establish invariance of $P(Y \mid X)$. For $m1$, the sparsity score is the number of non-zero entries in the selected feature block. Within each class, low- and high-score extremes supply the exact-size OOD set; training and ID samples are stratified draws from the remainder.

For $m2$, the final training set is fixed first. A MaxAbs scaler and TruncatedSVD representation are fitted on training data only, and a centroid is computed for each class. Every non-training sample is scored by Euclidean distance to its own-class centroid. Within-class low- and high-distance extremes form S_{OOD}, and the ID holdout is sampled from the remaining pool. This construction probes unusual geometry relative to the fixed training set without using test information to fit the representation.

4.4 Network-flow shift construction

Numeric preparation removes identifiers, explicit label derivatives, and categorical protocol fields, leaving 39 UNSW-NB15 and 89 ToN-IoT features. A leakage audit records every exclusion and feature–label association; the largest observed absolute correlation is 0.62 and belongs to a legitimate traffic feature. Each scenario has reference and current pools and uses two mechanisms.

The *m2-centroid* mechanism is the network analogue of EMBER $m2$: train-only MaxAbs+TruncatedSVD class centroids define within-class distance tails, which supply the OOD set. The *attack-campaign* mechanism performs no geometric selection. Training and ID pools contain benign traffic and all non-target attack categories; OOD contains benign traffic and the held-out target campaign. UNSW-NB15 reference and OOD pools additionally use different official capture partitions. This is an unseen-campaign plus capture-partition shift, not a timestamped temporal split.

4.5 Shared preprocessing and representations

For every run and dimension $d \in \{4, 6, 8, 10, 12\}$, a MaxAbs scaler and TruncatedSVD are fitted on T; the reduced coordinates are standardized and mapped linearly to $[0, \pi]$ with training-fitted parameters. The same embedded samples are supplied to both families, making the comparison a kernel swap rather than a representation swap. Dimension is a candidate axis. No validation, ID-test, or OOD-test row fits a transformation. The SVD random state equals the model seed.

4.6 Classifiers and train-only regularization

The primary learner is an SVC with a precomputed Gram matrix and balanced class weights [40]. For every kernel geometry, $C \in \{0.01, 0.1, 1, 10, 100\}$ is selected by five-fold stratified cross-validation on the training Gram matrix; ties favour the smaller C. When $n_{\text{train}} > 2000$, this selection step uses a class-stratified random-state-20260717 subsample capped at 2,000 rows, while the final model is always refitted on the complete training block. Because C is resolved before family selection, its grid points are not counted as separate family candidates.

The secondary learner is a binary Gaussian-process classifier with a logistic link and Laplace approximation [41]. It consumes the same precomputed kernel, uses diagonal jitter 10^{-8}, at most 100 Newton iterations, and convergence tolerance 10^{-6}. The logistic–probit approximation supplies predictive probabilities; no post-hoc calibration is applied.

4.7 Classical kernel family

The customary reference contains a cosine-normalized linear kernel and RBF kernel. The extended family adds cosine-normalized degree-2 and degree-3 polynomial kernels with $\gamma = 1/d$ and intercept 1, the Laplacian kernel $\exp(-\|x-x'\|_1/\ell_1)$, and Matérn-3/2 and Matérn-5/2 kernels. The base ℓ_1 and ℓ_2 scales are median pairwise Manhattan and

Euclidean distances, respectively, calculated from at most 2,000 training rows with the model seed; the RBF base scale is fitted from training variance. RBF, Laplacian, and both Matérn kernels use factors $\{0.1, 0.3, 1, 3, 10\}$ around these training-only bases. Deduplication yields 23 kernel-shape/scale blocks crossed with five dimensions, or 115 extended-classical candidates in full-coverage groups.

4.8 Quantum fidelity-kernel family

The quantum family uses exact statevector fidelities

$$K_Q(x, x') = |\langle \phi_Q(x) \mid \phi_Q(x') \rangle|^2$$

[1, 2]. Four feature-map blocks are evaluated: full-entanglement ZZ maps with one and two repetitions, a full-entanglement Pauli map with one repetition and Pauli strings X and Z, and a Z map with two repetitions. Each map uses angle-scale factors $\{0.5, 1, 2\}$ after the shared $[0, \pi]$ mapping and all five dimensions, giving 60 candidates. The dimensions correspond to 4–12 simulated qubits. Classical bandwidth and quantum angle-scale freedom serve the symmetric purpose of controlling locality [27, 28]; no circuit is trained.

4.9 Target-label-free family selection

The original ID holdout is divided deterministically and class-stratifiably into V_{ID} and S_{ID}. Within each class, global row indices are ordered by SHA-256 of the salt `ksf-v4-idsplit::` and index, then assigned alternately to validation and test. The halves are disjoint and approximately class-balanced. P1$'$ deploys, within each family, the candidate with highest V_{ID} balanced accuracy and reports it once on S_{ID} and S_{OOD}. Ties follow lexicographic candidate order. OOD labels never enter regularization, preprocessing, or P1$'$ selection.

4.10 Candidate-budget matching

Best-of-family selection rewards a larger pool. The primary comparison therefore matches the quantum budget: 60 candidates in full-coverage groups and 20 in reduced-coverage groups. For each group and classifier, 5,000 extended-classical subsamples are generated with seed 20260715. The primary `kernel_blocked` scheme samples complete five-dimension blocks—12 of 23 blocks for budget 60 and 4 of 7 for budget 20—to mirror the quantum feature-map/scale-by-dimension structure. `uniform` samples individual candidates, while `kernel_stratified` distributes the draw across kernel-shape/scale strata. Each subsample selects its P1′ winner; the classical endpoint is the resampling expectation. The full 115-candidate pool is secondary. Resampling percentiles describe sensitivity to the sampled candidate set, not population uncertainty.

4.11 Security fixed-case aggregation and conditional intervals

The effective replication axis within a scenario-group is q-split seed. Master seed is a design stratum rather than a replicate, some EMBER OOD pools recur across master seeds, and training sizes are nested. Audited OOD overlap is at most 1.2% for EMBER and 7.8% for network groups. Within each q-split cluster, model seeds are averaged within master seed, master seeds within size, and the three sizes with equal weight. The five resulting cluster means T_j give the group effect $\bar{T}$ and 95% conditional cluster-t interval

$$\bar{T} \;\pm\; t_{4,0.975}\frac{s_T}{\sqrt{5}}.$$

This interval ($n = 5$ q-split clusters) is conditional on the fixed benchmark pools and experimental design. A 9,999-draw BCa bootstrap over cluster means and leave-one-q-split calculations are secondary diagnostics.

Across groups, EMBER $m1/m2$ are averaged within source dataset, the four UNSW-NB15 groups within UNSW-NB15, and the two ToN-IoT groups within

ToN-IoT. These three source-dataset means receive equal weight. Leave-one-source-dataset-out ranges omit one of the three at a time. The eight scenario-groups and three datasets were not sampled from defined populations; therefore no population p-value or dataset-population confidence interval is reported.

4.12 Specification curve

The specification curve is a descriptive multiverse over every logically valid endpoint with complete paired legacy and v4 data. Its ten prespecified rows cross, where available, legacy or corrected generation; same-test OOD-oracle, legacy ID-based, or disjoint ID-validation selection; fixed $C = 1$ or train-only cross-validation for SVC; customary linear/RBF or extended references; and native or equal budgets. S1–S10 were ordered before aggregation and are never sorted by effect. Realizations are averaged within setting and scenario-group, then groups within the three source datasets, and finally source datasets with equal weight. All eight group effects remain visible and no specification-level hypothesis test is performed.

4.13 Prospectively frozen external corroboration

Before acquiring TableShift data or inspecting any external result, we committed the tasks, sampling, preprocessing, candidate pools, estimands, uncertainty procedure, and outcome-category rule. The source implementation is pinned to commit `fca9429814703a07e3902d005d46563a207b7f0a` [42]. Three public tasks span distinct domains and published shift variables: College Scorecard (institution type), diabetes readmission (admission source), and ACS income (geographic region). TableShift's `train`, `validation`, `id_test`, and `ood_test` roles are preserved; `ood_validation` is never loaded. College Scorecard is read from its public CC0 archive because the pinned downloader invokes authenticated Kaggle access; the pinned parser, schema, and domain split are unchanged. Source-file SHA-256 hashes and the container digest are recorded.

Each task has nested strata $(n_{\text{train}}, n_{\text{val}}, n_{\text{ID}}, n_{\text{OOD}}) = (1000, 250, 250, 500)$ and $(2000, 500, 500, 1000)$. For seeds $\{42, 123, 999, 7, 2024\}$, source rows are ordered by SHA-256 of task, split, seed, and original position, giving deterministic label-blind samples. Gates verify source-role disjointness, size nesting, both classes, source-schema preservation, and at least 12 non-constant encoded features. All 30 task–size–seed units passed.

All transformations fit `train` only. Constant columns are removed; numeric variables receive median imputation; Boolean, categorical, and string variables receive most-frequent imputation and one-hot encoding with unknown categories ignored. MaxAbs scaling, TruncatedSVD at the five study dimensions, standardization, and $[0, \pi]$ range mapping follow. The SVD random state is 20260729. The security kernel and classifier pools are then reused.

For each task, size, seed, and classifier, the validation-selected 60-candidate quantum endpoint is compared with 5,000 `kernel_blocked` 60-candidate subsamples of the 115-candidate classical pool using seed 20260729. The primary effect is quantum minus expected classical OOD balanced accuracy. The equal-budget OOD oracle selects on `ood_test`; the weak-fixed oracle uses linear/RBF, fixes SVC at $C = 1$, and also selects on the reported OOD labels. Protocol contraction is weak-fixed-oracle effect minus controlled P1′ effect.

Seeds are averaged within task and size, nested sizes receive equal weight within task, and the three fixed tasks receive equal weight. Conditional 95% intervals use 9,999 resamples of the five seed clusters within task while preserving nested sizes. Leave-one-task-out ranges are descriptive. The intervals quantify deterministic-subsampling variability conditional on three fixed tasks; they do not represent a population of tasks.

4.14 Performance and uncertainty metrics

Balanced accuracy on S_{OOD} is primary. Accuracy, macro and positive-class F1, ROC AUC, and precision–recall AUC are recorded on every split. The GP analysis additionally records log loss, Brier score, expected calibration error with 15 equal-width probability bins [43], and mean predictive entropy. No threshold or calibration parameter is fitted on OOD labels.

4.15 Geometry descriptors and association analyses

For positive-semidefinite K with eigenvalues λ_i and $p_i = \lambda_i / \sum_j \lambda_j$, effective rank is

$$r_{\text{eff}} = \exp\left(-\sum_i p_i \log p_i\right)$$

[22]. For $y \in \{-1, +1\}^n$ and doubly centered K_c, centered kernel-target alignment is

$$\text{KTA}(K, y) = \frac{\langle K_c, yy^\top \rangle_F}{n \|K_c\|_F}$$

[23, 24]. KTA is computed separately on training, ID, and OOD square blocks. We also record off-diagonal mean and standard deviation, cross-split similarities, and regularized geometric difference

$$g(K_C \to K_Q) = \left\| \sqrt{K_Q}(K_C + \lambda n I)^{-1} \sqrt{K_Q} \right\|_\infty^{1/2}$$

[4].

Within each run–dimension cell, Spearman correlations relate effective rank, OOD KTA, and KTA survival (OOD minus ID) to SVC OOD balanced accuracy for all, classical-only, and quantum-only candidates. Partial Spearman association residualizes ranked variables against family and log length/angle scale. For portability, a descriptive model bins log effective rank into ten training-dataset quantiles, learns mean OOD accuracy per bin from two source datasets, and ranks cells in the third; held-out Spearman correlation is reported once per source dataset.

Rank matching is observational. Within each run and dimension, every quantum candidate is first paired to the classical candidate with smallest absolute effective-rank difference; classical candidates may be reused. Rank ratio $\max(r_Q, r_C)/\min(r_Q, r_C)$ is filtered at prespecified calipers 1.10, 1.25, 1.50, and 2.00, plus unfiltered reporting. A sensitivity uses minimum-cost one-to-one assignment with absolute log-rank difference, leaving excess candidates unmatched. Counts, retained fractions, rank discrepancies, mean and median OOD differences, and the fraction favouring quantum are reported for every caliper. None of these analyses identifies a causal geometry or family effect.

4.16 Repeated finite-shot perturbation

The v0.6.0 finite-shot analysis was frozen before outcome computation. It fixes one $q1000$, master-seed-42, q-split-42, model-seed-42 run per security group and, within each, the exact-statevector quantum SVC candidate selected by P1′ on V_{ID}. Lexicographic candidate order resolves ties. Input `summary_v4.csv` SHA-256 hashes, chosen kernels, dimensions, and exact C values are stored in the frozen specification.

For shots $s \in \{128, 512, 2048, 8192\}$ and measurement replicates $r = 0, \ldots, 29$, each exact fidelity p is sampled independently as $\mathrm{Binomial}(s, p)/s$. A separate NumPy generator is used for train, ID-validation, ID-test, OOD-test, and OOD-square blocks. Its unsigned 64-bit seed is the first eight SHA-256 bytes of

```
ksf-v6-shots::<run>::<kernel>::<dim>::<shots>::<replicate>::<block>.
```

Python's process-randomized hash is not used.

Square train and OOD blocks are sampled on the upper triangle, mirrored, clipped to $[0, 1]$, and assigned unit diagonal. Both this pre-projection matrix and its PSD projection by negative-eigenvalue clipping are retained. Rectangular blocks receive binomial sampling and clipping only. For both projection conditions, C is reselected by the same five-fold train-only procedure and the final SVC is refitted. Outputs include signed and absolute OOD change from the fixed exact endpoint, effective-rank ratio, OOD-KTA change, negative-eigenvalue fraction, and separate Frobenius changes due to sampling and projection. Within each fixed Gram matrix, medians and central 2.5–97.5% Monte Carlo intervals summarize 30 replicates. Across the eight fixed matrices, medians and ranges are descriptive. This model contains no device or correlated-noise process.

4.17 Statistics and reproducibility

No null-hypothesis test is used for the eight security groups, three source datasets, or three external tasks. Every reported uncertainty object names its effective n, resampling unit, and conditioning set: security group intervals use five q-split clusters; external intervals use five deterministic seed clusters per task; finite-shot intervals use 30 measurement replicates conditional on one fixed Gram matrix. The fifteen security pipeline realizations are never treated as independent. Small-n security intervals use the t_4 reference and descriptive ranges accompany cross-dataset summaries. No multiplicity adjustment is required because no family of p-values is produced.

Sample sizes were design- and compute-constrained and fixed before outcome analysis; no power calculation was used. No completed group, task, specification, budget scheme, caliper, or finite-shot replicate was removed based on its result. Randomized operations use recorded seeds; external sampling and ID division use deterministic

SHA-256 ordering. The test suite validates source-dataset mapping, endpoint equality across analyses, candidate-budget coverage, matching uniqueness, seed stability, exact-reference hashes, and complete Monte Carlo cells.

4.18 Code availability and software

Experiments use Python 3.11/3.12, NumPy 2.3.5, pandas 2.2.3, scikit-learn 1.8.0, Matplotlib 3.11.0, and Qiskit 2.3.1. The exact cross-platform environment is pinned. Code, manifests, tests, aggregated results, and figure/table generators are available at https://github.com/roberto-fernandez-barrios/kernel_shift_framework. The exact v0.6.0 submission release is identified by the reserved immutable DOI https://doi.org/10.5281/zenodo.21672470; the concept DOI resolves all versions.

4.19 Use of generative AI

OpenAI Codex was used during manuscript revision for language editing, code inspection, analysis implementation, and consistency checking. It did not determine authorship, study accountability, or acceptance of any scientific claim. The authors inspected the generated changes, verified all reported analyses and citations, and take full responsibility for the manuscript.

Declarations

Data Availability. This study uses the public EMBER [13], UNSW-NB15 [16], ToN-IoT [17], and TableShift [42] benchmarks, which remain subject to their original access conditions. Raw source records are not redistributed. Derived split definitions, deterministic row-position hashes, shift constructions, experiment manifests, complete aggregated results, and summary tables are available through the project Zenodo concept record at https://doi.org/10.5281/zenodo.19147649. The exact v0.6.0 submission artifact is identified by the reserved immutable version DOI https://doi.org/10.5281/zenodo.21672470.

Code Availability. Code for constructing the splits, running the kernel experiments, reproducing every analysis, and regenerating every table and figure is available at https://github.com/roberto-fernandez-barrios/kernel_shift_framework and in the versioned Zenodo archive above. The platform-independent environment is pinned in `environment.lock.yml`; the code is released under BSD-3-Clause with no restriction on non-academic use.

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Authors' contributions. R.F.-B.: conceptualization, methodology, software, data curation, formal analysis, visualization, investigation, writing—original draft, and writing—review and editing. I.P.-L.: supervision, validation, and writing—review and editing. A.G.-S.: validation, project administration, and writing—review and editing. P.G.B.: supervision, resources, and writing—review and editing. All authors read and approved the final manuscript.

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